

Tech for Good vs. Tech for Profit

Ethical Conflicts in Startups

Engineering Ethics

Hum 4441

Group No – 1

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Overview

**Focus on bKash and
Nagad as case
studies in
Bangladesh**

**Use of mobile
phones for financial
transactions,
including among the
rural poor**

**Combine financial
inclusion goals with
profit-driven
innovation**

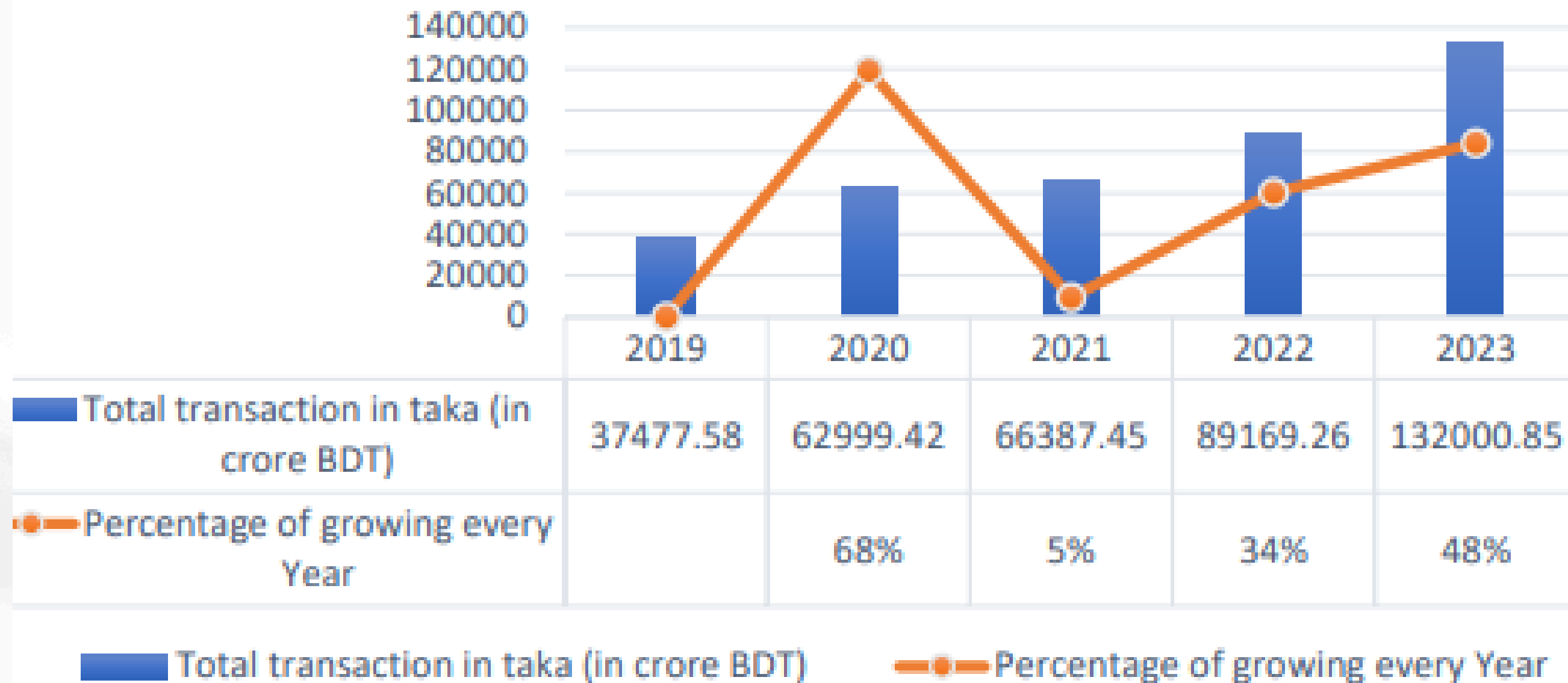
Importance of Ethics in FinTech

- MFS systems handle sensitive data and public money; errors have real-world impact
- Ethical design ensures fairness, inclusion, and protection for vulnerable users
- Regulatory compliance builds trust and prevents financial misuse
- Frameworks like VSD and IEEE guide responsible, human-centered tech development
- Ethics is essential not just for safety but for long-term success

Global & Ethical Relevance

- Bangladesh is a global leader in digital inclusion (910 MFS accounts per 1,000 adults)
- UN & IEEE frameworks stress user protection, gender equity, and transparency
- Poorly governed fintech can worsen inequality and create systemic risks
- The bKash–Nagad contrast shows how ethics affect scale, trust, and sustainability
- Ethical fintech design is a global necessity, not just a local concern

MFS transactions amount and YoY percentage growth



Source: Bangladesh Bank



Ethical Dilemma

Balancing financial inclusion and innovation with profit motives and regulatory compliance in mobile financial services (MFS)

Can fintech firms scale rapidly while ensuring user trust, transparency, and fairness, especially for marginalized users

Core Issue



Affected Stakeholders



General users



MFS companies



Government regulators



Government regulators



Agents and small
merchants

Objectives

- Explore ethical challenges in MFS through case studies
- Apply global frameworks (VSD, IEEE, UN, Responsible Innovation)
- Analyze how values like **trust and transparency** shape design
- Propose an ethical-operational model **balancing impact and profit**

Literature Review – 1

IEEE Ethically Aligned Design:

A Vision for Prioritizing Human Well-being with Autonomous and Intelligent Systems

Key Concerns: Risks of harm, lack of accountability

Takeaways: Promote human rights, transparency, and safe tech



Literature Review – 2

UN Principles for Responsible Digital Payments (Better Than Cash Alliance, 2021)

Key Concerns: User exploitation, lack of fee clarity

Takeaways: Ensure fund protection, fairness, and inclusion



Literature Review – 3

Value-Sensitive Design: Theory and Methods (Friedman & Khan)

Key Concerns: Ignoring user values in system design

Takeaways: Embed trust, privacy, and inclusion into tech design

Literature Review – 4

Responsible Innovation: A Framework for Anticipating Societal Impact (UKRI, 2021)

Key Concerns: Unintended harms from unchecked innovation

Takeaways: Plan for inclusion, foresight, and ethical risks

Literature Review – 5

Fintech and Human Rights: Managing Social Risks to Maximize Benefits (BSR, 2025)

Key Concerns: Data misuse, bias, and misleading marketing

Takeaways: Governance must safeguard user rights and trust

Literature Review – 6

Fintech for Financial Inclusion in Bangladesh (IMF, 2022)

Key Concerns: Exclusion of rural/women users, literacy gaps

Takeaways: Target marginalized groups with inclusive design

Literature Review – 7

ACM Code of Ethics and Professional Conduct (2018)

Key Concerns: Harm, dishonesty, unfair systems

Takeaways: Be transparent, fair, and socially responsible



*Committee on
Professional Ethics*

Literature Review – 8

**“Drivers Influencing the Adoption Intention towards Mobile Fintech Services: A Study on the Emerging Bangladesh Market”
(MDPI Information, 2022)**

Key Concerns: Overreliance on agent interactions, trust as a fragile foundation, low awareness of risk

Takeaways: Design beyond UI, highlight perceived benefits, leverage social influence

Literature Review – 9

Measuring the True Cost of Digital Services Innovations for Poverty Action (IPA), 2023

Key Concerns: Hidden fees, agent-level info gaps

Takeaways: Better agent training and fee transparency needed

Literature Review – 10

Financial Inclusion in Bangladesh: Challenges & Opportunities Bangladesh Bank & UNDP, 2021

Key Concerns: Limited rural access, low digital literacy

Takeaways: Use local agents, multi-language, low-tech tools

METHODOLOGY

Research Approach:

- Qualitative, interpretive case-study method
- Data from 2023–2025: company reports, news, NGO analyses
- Mixed-method triangulation: quantitative metrics + qualitative insights

Ethical Frameworks for Analysis:

- **Value-Sensitive Design (VSD):** Focus on user values like fairness, access and privacy
- **Responsible Innovation (UKRI):** Anticipate impacts, avoid harm, include diverse stakeholders
- **IEEE Ethically Aligned Design:** Uphold human rights, transparency, accountability
- **UN Digital Finance Principles:** Emphasize user protection, gender equity, and fund security

Case Study Selection Method

Purposive sampling: bKash and Nagad selected as dominant MFS platforms in Bangladesh

Chosen for contrast: one development-driven (bKash), one growth-driven (Nagad)

Both cases offer rich insights into real-world ethical tensions in fintech

Case Study – 1

Nagad's Use of Customer Deposits for Loans

Source: Business Inspection, 2024

Summary: Nagad used BDT 5B of user funds as loan collateral, violating Bangladesh Bank rules. When the loan defaulted, BDT 3.17B in deposits were used to repay.

Ethical Concerns & Impact

Fund Safety: Violated user trust by misusing deposits

Transparency: Users unaware of risks; no clear disclosure

Accountability: No penalties; undermines regulatory credibility

Case Study – 2

Nagad's Unlicensed Operations and Data Privacy Issues

Source: BBC Bangla, TBS, 2023

Summary: Nagad operated without a full MFS license, auto-created user accounts without consent, and faced allegations of data deletion.

Ethical Concerns & Impact

Privacy Violation: Accounts created without user

permission **Regulatory Evasion:** Undermines fair industry practices

Erosion of Trust: Weakens user confidence in digital finance

Case Study – 3

Growth Paths: Ethical Compliance vs. Regulatory Shortcuts

Source: Industry Insider, Gulf News, Business Inspection (2023–2025)

Summary: BKash expanded with full regulatory compliance and social programs; Nagad scaled rapidly using aggressive tactics, bypassing licensing norm

Ethical Concerns & Impact

BKash: Followed proper licensing, avoided scandals → built long-term trust

Nagad: Operated without full license, took risky financial shortcuts → created systemic risk

Impact: Ethical compliance enabled bKash's stable growth; Nagad's shortcuts led to regulatory backlash

Case Study – 4

Transparency in Pricing and User Awareness

Source: IPA Study, bKash website, Media Reports

Summary: BKash publishes fee charts online, but only ~4% of agents disclose fees. Nagad offers zero-cost promotions but lacks online pricing transparency.

Ethical Concerns & Impact

BKash: Transparency in principle, but weak agent-level enforcement

Nagad: Low upfront cost, but opaque fee structure → user confusion

Impact: Lack of clear pricing harms informed user choice and undermines trust

Case Study – 5

Tech-for-Good Initiatives vs. Profit-Centric Incentives

Source: bKash Blog, FFNews, Gulf News

Summary: BKash invests in rural literacy and inclusive services. Nagad focuses on cashback offers, free onboarding, and expensive promotions to grow fast

Ethical Concerns & Impact

BKash: Values-based outreach → empowers underserved communities

Nagad: Incentive-driven growth → risks short-termism and misuse of funds

Impact: : bKash's inclusive model aligns with ethical design; Nagad's approach raises sustainability concerns

Comparative Overview

Aspect	bKash	Nagad
Launch / Ownership	2011; BRAC Bank, IFC, Gates Foundation, etc.	2019; Bangladesh Post Office (51%) + Third Wave (49%)
Users (2023)	~74 million	~80 million (claimed >95M)
Agent / Outlet Network	364,165 agents; strong NGO links	10,000 post offices + Uddokta agents
Revenue / Profit	BDT 41.9B revenue; Tk 1.04B profit (504% YoY growth)	Not disclosed; unicorn valuation; reportedly large financial losses
Social Initiatives	Digital literacy programs, utility bill pay, NGO partnerships	Govt. subsidy disbursements, QR onboarding for 300K SMEs
Pricing Transparency	Fees listed online, but agents rarely explain (~4% disclosure)	No online fee info; reliant on promos and incentives
Tech Innovations	App + USSD, savings/credit, multi-language support	First full e-KYC; hybrid USSD-app; free QR for merchants
Regulatory Issues	Fully licensed; no major violations	Unlicensed operations, misuse of funds, data privacy concerns

Ethical Issues and Analysis

Main Ethical Issues

- **Privacy:**
 - Sensitive data, identity theft, profiling, unauthorized surveillance.
- **Security:**
 - Threats: phishing, SIM swap, server vulnerabilities.
- **Fairness:**
 - Issues affecting rural or low-literacy users, non transparent pricing that burdens marginalized groups.
- **Transparency:**
 - Users should know where and how their money is being used
- **Accountability:**
 - In case of transactions frauds, responsible actions must be undertaken
- **Inclusion Metrics:**
 - Policies to motivate the marginalized community

Ethical Viewpoints



Utilitarianism



Deontological
Ethics



Virtue Ethics



Rights-Based
Ethics



Justice Ethics

Proposed Ethical Framework

- **Stakeholder-Centered Design:** Perform design sprints; ensure multi-channel (app, USSD, voice) and local-language support.
- **Dual-Inclusion Metrics:** Track both financial (revenue, ARPU) and social KPIs (rural adoption, women's usage, literacy gains); tie management incentives to both.
- **Radical Transparency & Fairness:** Publish fees everywhere (app, website, agent outlets); train agents on disclosure; fully segregate customer funds; provide clear recourse channels.
- **Tech-for-Good Features:** Embed in-app literacy tips and fraud warnings; offer cross-subsidized, tiered pricing for low-balance users; partner with NGOs/government to reach real needs.

- **Governance & Accountability:** Mandate independent financial audits; convene ethics advisory boards (advocates, regulators, experts); report inclusion progress publicly.
- **Iterative Responsiveness:** Use analytics and user feedback to spot and fix emerging harms; run regular stress-tests; proactively engage regulators throughout development.
- **Balanced Profit Model:** Allocate a share of profits to community initiatives or maintain low fees via cross-subsidies—aligning commercial success with social impact.

Policy & Regulatory Recommendations

- Mandate Inclusion-Linked Licensing
- Enforce statutory segregation of customer funds, with quarterly certified audits made public
- Create an independent body to handle end-user complaint
- Incentivize Responsible Innovation(that embed VSD/IEEE-aligned processes etc)

Ethical System Design & Governance

- Form Advisory committee—including consumer advocates, ethicists, and regulators—to vet major product changes and promotional campaigns.
- Impact-Tied Incentives with inclusion and trust metrics (e.g., agent disclosure rates, customer satisfaction scores)
- Publish biannual “Ethics & Impact” report
- Cross-Subsidy Pricing Models

Engineers' Key Responsibilities

- **Deep User Empathy**
 - IEEE Ethically Aligned Design (EAD)
 - Value-Sensitive Design (VSD)
 - ACM Code of Ethics
- **Bulletproof Transaction Integrity**
 - PCI DSS (Payment Card Industry Data Security Standard)
 - UN Principles for Responsible Digital Payments
- **Total Transparency Engine**
 - IEEE Ethically Aligned Design (EAD)
 - IEEE Code of Ethics
 - ACM Code of Ethics
- **Relentless Quality & Security**
 - IEEE Code of Ethics
 - PCI DSS (Payment Card Industry Data Security Standard)

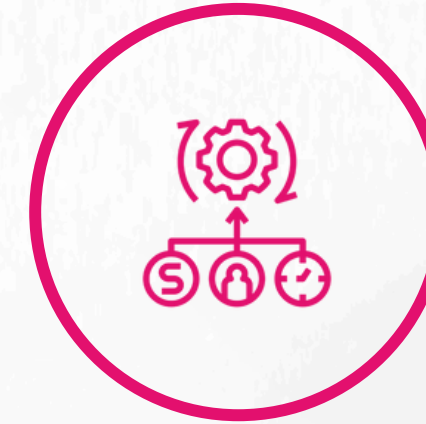
Challenges



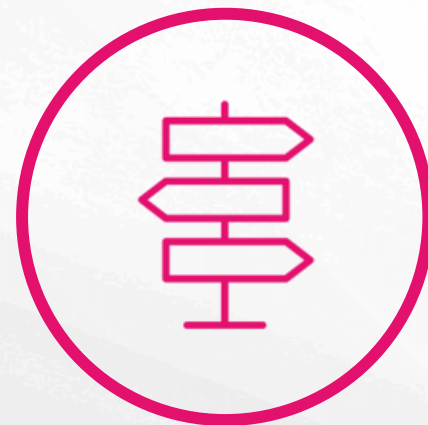
Agent Information
Gaps



Digital Divide



Training Resource Limits



Regulatory Ambiguity



Legacy IT Conflicts

Future Direction

Focus on Ethical AI, Cybersecurity & Data Rights

- **Responsible AI Governance**
- **Zero-Trust Cybersecurity**

Evolving Role of Ethics in IT

- **From Compliance to Culture**
- **Dynamic Ethical Frameworks**

Encouraging Interdisciplinary Collaboration

- **Cross-Functional Teams**
- **Public-Private-Academic Consortia**
- **Community Co-Design Networks**

Conclusion

Synthesis of Key Findings

- **Inclusion vs. Profit Balance:** bKash and Nagad prove that MFS can expand financial access, but their contrasting paths show the tension between scale and ethical governance.
- **bKash:** Profit-driven reinvestment in rural outreach, fee transparency, and multi-channel design demonstrated how "tech for profit" can support "tech for good."
- **Nagad:** Growth via e-KYC and zero-fee strategies brought scale but exposed risks—unlicensed ops, deposit misuse, and privacy gaps—highlighting the dangers of ethics sidelining.

Conclusion

Call for Proactive Ethical Practices

- **All stakeholders—startups, investors, regulators, communities—must co-create systems that prioritize ethics through:**
 - **Regulatory safeguards** (e.g., escrow segregation, licensing tied to inclusion)
 - **Internal structures** (ethics boards, participatory design)
 - **Investments** in literacy, secure engineering, AI bias checks, and user data ownership

Thank You!

Appendix

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